Safety Framework Release Notes

AVR SD Safety Framework 1.0.0

What is the Safety Framework?

The AVR® SD Safety Framework is an ISO 26262 ASIL C and IEC 61508 SIL 2 compliant software framework written in C99 that coordinates the hardware safety mechanisms of the AVR SD family. From the first instructions after reset, it initializes and diagnoses the safety system, schedules latent fault diagnostics alongside the application under watchdog supervision, centralizes software and hardware error handling and manages Safe State transitions and reset recovery. It also provides the execution flow for the application, so development starts from a working safety system and can focus on the application-specific functionality.

Packages

The Safety Framework is available in three packages. The Public Documentation and the Source Package are free of charge and contain everything needed to evaluate the framework and develop with it. The Functional Safety Package adds the safety, compliance and verification documentation needed to use the framework in a safety application. The table shows the contents of each package:

Content	Public Documentation	Source Package	Functional Safety Package
API documentation and UML diagrams	Yes	Yes, with source code	Yes, with source code and unit tests
Release Notes, release manifest and checksum file	Yes	Yes	Yes
Framework source code		Yes	Yes
MPLAB® X example project		Yes	Yes
Software Safety Manual			Yes
Software FMEA and DFA			Yes
MISRA C:2025 Compliance Plan and Report, HIS Metrics Statement			Yes
ISO 26262, IEC 61508 and ASPICE Process Assessment Result Summaries			Yes
Unit test source, test results and code coverage reports			Yes
Static analysis, code metrics, compiler diagnostics and SBOM reports			Yes
Clang-Tidy and Clang-Format configuration files			Yes

The README in each package describes its contents in more detail.

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1. Download

- **Public Documentation**: free of charge in the public GitHub repository
- **Source Package**: free of charge, with immediate access through the source access form on microchip.com
- **Functional Safety Package**: available on Microchip Direct as AVR-SD-SAFETY-FRAMEWORK

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2. Operating Environment

The delivered verification results apply to the following toolchain:

- MPLAB® XC8 Functional Safety Compiler v3.19
- Device Family Pack AVR-Sx_DFP 1.2.26

For use in a safety application, the framework must be compiled with this compiler and Device Family Pack version. Other MPLAB® XC8 compilers and Device Family Pack versions can be used for evaluation. Using a different version in a safety application requires an assessment of the impact on the Safety Measures, as described in the Software Safety Manual.

The example project in the Source Package and the Functional Safety Package requires MPLAB® X IDE v6.30 or later. The framework itself does not depend on an IDE.

The framework is not designed to be compiled with zero optimization (`-O0`). See *Restrictions* on the *Home* page of the API documentation for details and for how to debug individual files at `-O0`.

2.1. Supported Devices and Families

The Safety Framework supports the following AVR® SD devices:

- AVR32SD20
- AVR32SD28
- AVR32SD32

2.2. Memory Footprint

Measured with all optional features enabled, compiled at `-Os` for the AVR32SD32 with the compiler and Device Family Pack listed above.

Resource	Footprint
Flash	Less than 14.5 KB
Static SRAM	Less than 50 bytes
Dynamic memory allocation	None

The footprint of an application depends on the enabled features and the application code. The estimated worst-case execution times of the framework Tasks and diagnostics and the maximum stack depth are documented on the *Measurements* page of the API documentation.

2.3. Related Hardware and Documentation Support

Title	Location
AVR® SD Family	https://www.microchip.com/en-us/products/microcontrollers/8-bit-mcus/avr-mcus/avr-sd
AVR32SD32 Curiosity Nano	https://www.microchip.com/en-us/development-tool/ev75s16a
Safety Framework API documentation	https://microchip-pic-avr-solutions.github.io/avr-safety-framework/
AVR-Sx Device Family Pack	packs.download.microchip.com/#pack-Microchip-AVR-Sx-DFP-pdsc
Download Free MPLAB® XC Functional Safety Compilers	www.microchip.com/en-us/tools-resources/develop/mplab-xc-compilers/functional-safety
Purchase MPLAB® XC8 Functional Safety Compiler v3.19 certification package	www.microchipdirect.com/dev-tools/SW-006021-FS3-19
AVR® SD Device Functional Safety Package (ISO 26262)	https://www.microchipdirect.com/product/FBP-AVR-SD-ISO26262
AVR® SD Device Functional Safety Package (IEC 61508)	https://www.microchipdirect.com/product/FBP-AVR-SD-IEC61508

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3. Getting Started

1. Read the Safety Framework API documentation, starting with the *Home* page and *Software Architecture*.
2. Run the example project, included in the Source Package and the Functional Safety Package. See `examples/safety_framework_default_example.X/README.md` for how to run it.
3. Follow *Getting Started* on the *Home* page to adapt the framework to the application. Its *Development Tips* cover watchdog timing, debugging with the example callbacks and adding a first Task that shows the heartbeat on the Curiosity Nano LED.
4. Before using the framework in a safety application, read the Software Safety Manual and the Software FMEA in the Functional Safety Package.

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4. What's New?

1.0.0 - Initial Release

- Initial release of the AVR® SD Safety Framework.

4.1. Repairs and Enhancements

Not applicable for the initial release.

4.2. Known Issues

None.

Issues and documentation errata identified after this release are published in the Known Issues List, which is maintained separately from the release packages.

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5. Verifying the Package

Each package contains `release_content.sha256`, which lists the SHA-256 checksums of the release files. To verify the package, run the following command from the package root:

```
sha256sum -c release_content.sha256      # Linux, WSL or Git Bash
shasum -a 256 -c release_content.sha256  # macOS
```

The Functional Safety Package also has a checksum file for the ZIP archive itself. See the README in the package for details, including verification on Windows and Git line ending settings.

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